# tf.compat.v1.math.atanh Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/atanh

> *Computes inverse hyperbolic tangent of x element-wise.*

## Function Signatures

```
tf.compat.v1.math.atanh( x: Annotated[Any, tf.raw_ops.Any], name=None ) -> Annotated[Any, tf.raw_ops.Any]
```

## Code Examples

```
tf.compat.v1.math.atanh(
x: Annotated[Any, tf.raw_ops.Any],
name=None
) -> Annotated[Any, tf.raw_ops.Any]
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```
tf.raw_ops.Any
```


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tf.raw_ops.Any
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```
x = tf.constant([-float("inf"), -1, -0.5, 1, 0, 0.5, 10,
float("inf")])
tf.math.atanh(x) ==> [nan -inf -0.54930615 inf  0. 0.54930615 nan
nan]
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x = tf.constant([-float("inf"), -1, -0.5, 1, 0, 0.5, 10,
float("inf")])
tf.math.atanh(x) ==> [nan -inf -0.54930615 inf  0. 0.54930615 nan
nan]
```

# Documentation

Computes inverse hyperbolic tangent of x element-wise.

Given an input tensor, this function computes inverse hyperbolic tangent
for every element in the tensor. Input range is [-1,1] and output range is
[-inf, inf]. If input is -1, output will be -inf and if the
input is 1, output will be inf. Values outside the range will have
nan as output.

## Returns